

Sreehar Sanisetty

DATA ENGINEER

USA | +1 551-465-6063 | sreeharsanisetty7@gmail.com | [LinkedIn](#)

SUMMARY

- Results-driven and detail-oriented **Data Engineer with 4+ years of experience** in architecting, building, and optimizing scalable data pipelines and analytics solutions in agile enterprise environments.
- Proficient in developing robust **ETL/ELT processes** using tools like **Apache Spark, Kafka, Airflow, dbt**, and cloud services including **AWS and Azure**.
- Strong expertise in **Python, SQL, and R**, with a solid foundation in data modeling, data warehousing, and real-time streaming. Adept at working with relational and **NoSQL databases such as MySQL, PostgreSQL, Oracle, and MongoDB**, ensuring high availability and data integrity.
- Skilled in data visualization and storytelling using **Tableau, Power BI, and Excel**, translating complex datasets into actionable insights for business stakeholders.
- Experience deploying machine learning models using **Scikit-learn, TensorFlow, and BERT/GPT-based NLP models**, with a keen focus on prompt engineering and AI integration into data pipelines.
- Hands-on experience with **big data platforms** (Hadoop, Beam), **containerization (Docker)**, and **CI/CD pipelines** using **GitHub, Jenkins, and Terraform**. Well-versed in **SDLC, Agile, and Waterfall** methodologies, with strong communication, documentation, and cross-functional collaboration skills.
- Committed to data privacy and compliance with frameworks like **HIPAA, PCI DSS, and RBAC**, incorporating **data masking, tokenization, and data lineage** practices. Known for delivering high-performance, production-ready solutions that enhance business intelligence and decision-making across domains.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Languages:	Python, SQL, R,
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
Visualization Tools:	Tableau, Power BI, Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Cloud Platforms:	AWS, Azure
Database:	MySQL, Oracle SQL, PL/SQL, PostgreSQL, MongoDB
Data Engineering Concepts:	Apache Spark, Apache Hadoop, Apache Kafka, Apache Beam, dbt, ETL/ELT, PySQL, PySpark
Generative AI:	GPT-4, BERT, LLaMA, LangGraph, RAG, Prompt Engineering, Tokenization
Other Technical Skills:	DAX, SAS, JIRA, SAP, SSIS, SSRS, Machine Learning Algorithms, Probability distributions, Hypothesis Testing, Regression Analysis, Advanced Analytics, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Storytelling, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, Ad Hoc Analysis, Project Management, Data Presentation, Requirement Gathering, Root Cause Analysis, Quantitative Analytics, Docker, Big Data & AI Integration, PCI DSS, Data Lineage & Masking, RBAC, CI/CD Pipeline
Version Control Tools:	Git, GitHub
Operating Systems:	Windows, Linux, Mac OS

+

EXPERIENCE

Data Engineer | Goldman Sachs, CT

Sept 2024 – Present

- Established and managed a robust data pipeline to ingest and process financial transaction and market data from diverse sources such as trading platforms, clearinghouses, and external data vendors, utilizing **Apache Kafka** and **Apache NiFi**.
- Engineered ETL processes to convert and load raw trading and risk data into a Hadoop-based data lake, employing **Apache Spark** for distributed processing and implementing automated validation checks to ensure data accuracy and regulatory compliance.

- Designed **data models in MongoDB** to capture complex nested documents, such as multi-asset portfolios and hierarchical risk metrics, supporting advanced financial analysis.
- Orchestrated Extract, Transform, Load (ETL) workflows from multiple source systems to **Azure Data Storage** services using **Azure Data Factory, Spark SQL**, with Azure Data Lake Analytics for scalable financial data processing.
- Modelled complex ETL workflows as stateful graphs using **Langraph**, with nodes representing data ingestion, transformation, validation, and storage tasks, improving pipeline observability and debugging.
- Fine-tuned data pipelines for optimal performance and scalability using partitioning, bucketing, and indexing strategies across data lakes and warehouses to handle high-frequency trading and market data streams.
- Optimized **MapReduce** job performance through efficient key-value design and combiners, reducing processing time by **25%** on peak trading day data loads.
- Processed unstructured financial texts and market news feeds using **BERT** embeddings, converting qualitative insights into structured formats ingested into the data lake for real-time market impact analysis.
- Developed and maintained automated data quality validation and reconciliation rules using **Apache Airflow** and custom Python scripts to ensure consistency across multiple financial datasets.
- Collaborated with DevOps teams to containerize and deploy data pipelines using **Docker**, ensuring reproducible and scalable deployments across development, staging, and production environments in a regulated financial infrastructure.

Data Engineer | Cognizant, India

May 2020 – Aug 2023

- Architected and implemented a high-performance **ETL pipeline using Apache Kafka and AWS Kinesis**, processing over 2TB of healthcare data daily with 99.9% reliability, ensuring efficient data flow and transformation.
- Developed and maintained scalable data transformation scripts using **Python, PySpark, and AWS Glue**, facilitating complex /data manipulations and integrations for advanced analytics on large-scale healthcare datasets.
- Designed comprehensive **ER diagrams** to model complex relationships between financial entities such as trades, portfolios, counterparties, and market instruments, ensuring clear data architecture.
- Achieved a **20% improvement** in prediction accuracy by implementing **machine learning models** using AWS SageMaker and scikit-learn, leveraging data-driven insights for enhanced healthcare decision-making.
- Designed and published interactive dashboards and detailed reports using **Tableau**, boosting stakeholder access to actionable healthcare insights by 50% and reducing report generation time by **15%**.
- Designed and implemented **star schema data models** to optimize complex financial analytics and reporting workflows, enabling fast aggregation of trade and risk data.
- Enabled **re-identification controls for authorized systems via token vaults**, allowing secure reverse mapping of tokens for care providers or auditors while maintaining strict access logs and encryption at rest.
- Developed predictive models using scikit-learn and AWS Machine Learning to forecast healthcare resource utilization, achieving 90% accuracy in predicting hospital admissions and emergency room visits.
- Utilized **Snowflake** for data warehousing, integrating and consolidating data from various healthcare sources, including EHRs and claims, for comprehensive **OLAP analytics**.
- Orchestrated data workflows using **Apache Airflow on Amazon EKS (Elastic Kubernetes Service)**, managing complex ETL processes and ensuring timely data updates across the healthcare analytics platform.
- Implemented infrastructure-as-code using Terraform to provision and manage AWS resources, reducing deployment time by **10%** and enhancing consistency across development, staging, and production environments.
- Designed a data model for healthcare analytics, optimizing SQL (Amazon RDS) database for OLTP and OLAP workloads, with efficient indexing.
- Implemented a hybrid data architecture combining **Hadoop and AWS services**, allowing seamless data processing and storage between on-premise and cloud environments for enhanced scalability and flexibility.
- Leveraged Git for version control and implemented a robust **CI/CD pipeline using Jenkins** and AWS CodePipeline, automating testing and deployment processes, resulting in **50%** faster release cycles for data engineering projects.

EDUCATION

Master of Science in Data Science - University of New Haven, New Haven, CT, USA

Bachelor of Technology in Electrical and Electronics Engineering - R.M.K Engineering College, Chennai, India